Vera: A Low-Cost Open-Source Single-Channel EEG Seizure Monitor and a Reproducible Benchmark of Its On-Device Detectors

Sharan Tangalapally
VNR Vignana Jyothi Institute of Engineering and Technology
Hyderabad, India
shwethasharan.office@gmail.com

Vedanti Renghe
Terna Medical College
Navi Mumbai, India
vedantimr@gmail.com

DRAFT — numbers are from a PILOT run (full CHB-MIT run, but v1 front-end: no 0.5–40 Hz filter, FP1-F7 instead of FP1-FP2; rerun notebook v2); regenerate numbers.tex from the full results before submission.

Abstract*—Consumer-grade biopotential amplifiers and Wi-Fi microcontrollers now make an electroencephalography (EEG) seizure monitor buildable for a few thousand rupees, but the detection algorithms shipped with such devices are rarely tested against expert-annotated recordings. We present *Vera*, an open-source wearable monitor built from an EXG BioAmp Pill front-end and an ESP32 that records one frontal channel (FP1–FP2), runs a seizure state machine on the device, and alerts a phone over its own Wi-Fi hotspot, with an optional multi-feature analyzer on a laptop. To measure what these detectors actually do, we replay the CHB-MIT Scalp EEG Database (24 cases, 191 seizures, 223 h) through a model of Vera's analog front-end and run the *unmodified* firmware logic and analyzer code against it under SzCORE-style event scoring with a chance-level baseline. On FP1-F7, the shipped analyzer detected 16.2% of seizures at 2.16 false alarms/h, and the firmware detector 80.6% at 25.75/h (chance at the same alarm rate: 63.5%). Of the analyzer's three features, only line-length carried seizure information (pooled AUC 0.65, median per patient 0.79). The delta/alpha ratio (0.45) and high-frequency energy (0.48) were no better than chance. A scale-invariant line-length-ratio detector small enough to run on the ESP32, tuned by patient-wise cross-validation, detected 32.5% of seizures in held-out patients at 0.93 false alarms/h. That is about twice the shipped analyzer's sensitivity at less than half its false-alarm rate, although the absolute numbers show how hard seizure alarms are from one frontal channel. All hardware, firmware and evaluation code is public.

***Index Terms*—epilepsy, seizure detection, wearable EEG, consumer health technology, ESP32, open hardware, CHB-MIT, reproducibility**

I. Introduction

Epilepsy affects about 50 million people worldwide [1]. Many seizures happen unwitnessed, particularly during sleep. Unwitnessed convulsive seizures are a major risk factor for sudden unexpected death in epilepsy (SUDEP) [2], and patients' own seizure diaries miss a large share of events [3]. Automated seizure alarms therefore have direct value for patients and caregivers. A clinical practice guideline from the ILAE and IFCN now recommends wearable detectors, but only for devices validated on generalized tonic–clonic seizures [4]. Commercial wearables mostly sense movement and electrodermal activity [5], so they cannot see non-convulsive seizures, which are visible only in the EEG.

Two trends make a consumer EEG alarm possible. Open biopotential front-ends now achieve signal quality comparable to clinical amplifiers for many tasks [6], [7], and Wi-Fi microcontrollers such as the ESP32 [8] can process a single channel in real time. A home-built monitor can therefore cost very little. The open question is whether its detection logic works: thresholds chosen by hand, "pre-ictal" warning states and adaptive baselines are easy to implement and hard to validate.

This paper makes four contributions:

1) **Vera**, an open, battery-powered single-channel EEG monitor (front-end, ESP32 firmware, desktop analyzer and Android companion app) that raises alarms on the device without any internet connection (Sec. III).
2) **A code-faithful benchmark.** The shipped detector code runs unmodified on the CHB-MIT database [9], [10], which is replayed through a model of Vera's analog front-end on its own FP1–FP2 montage. Scoring is event-based, with a chance-level baseline (Sec. IV).
3) **A feature-level analysis** showing which of the shipped features carry seizure information on a single frontal channel (Sec. V).
4) **An ESP32-feasible line-length-ratio detector**, evaluated only on held-out patients, that addresses the weaknesses the benchmark exposes.

II. Related Work

Automated seizure detection on scalp EEG dates back to Gotman's 1982 detector [11]; Baumgartner and Koren review the field [12]. Line-length, the sum of absolute sample-to-sample differences, was introduced as a cheap and effective onset feature by Esteller *et al.* [13]. On CHB-MIT, patient-specific classifiers that use all channels detect most seizures within seconds [14]. Reduced montages, such as behind-the-ear electrodes, keep useful but lower sensitivity [15]. How

results are reported matters as much as the detector. Event-based scoring with explicit onset and offset tolerances, and false alarms normalized per 24 h, are now recommended [16], [17], and wearables should be judged by both sensitivity and false-alarm rate [4]. We follow these recommendations. We also report the sensitivity a *random* alarm process with the same alarm rate would achieve, because a detector that alarms often enough will appear sensitive by chance.

III. System Design

A. Hardware

Fig. 1 shows the signal chain. A single differential channel between the left and right forehead (FP1–FP2) is amplified by the EXG BioAmp Pill [7], buffered by an LM358 follower and digitized by the ESP32 at about 200 Hz. Forehead electrodes are hair-free, which suits gel-free consumer use, but they pick up eye-movement and blink artifacts. The system has no galvanic isolation, so with electrodes on the skin it must be powered only from a battery bank. A GPIO line delivers a 500-ms pulse on seizure onset for an external buzzer or Bluetooth trigger, and an RGB LED shows the current state. The bill of materials costs [TODO: total BOM cost in INR/USD].

B. On-device detector (ESP32 firmware)

For each sample $x[n]$ (12-bit ADC code), the firmware tracks an exponential baseline $b[n] = 0.99\,b[n-1] + 0.01\,x[n]$ and the deviation $d[n] = |x[n] - b[n]|$. While the state is NORMAL, d updates an exponentially weighted Welford estimate of mean μ and spread σ with forgetting factor 0.995 per sample. After a 90-s warm-up, the BROWNOUT ("warning") and BLACKOUT ("seizure") thresholds are reset every 30 s to $\mu + 2.5\sigma$ and $\mu + 4\sigma$, clamped to fixed ranges. Leaky counters require 20 (BROWNOUT) or 15 (BLACKOUT) supra-threshold samples before a transition. BLACKOUT is followed by a POST-ICTAL hold of 400 samples. The firmware serves state, battery and a 128-sample waveform buffer over a REST API on its own Wi-Fi access point, so no router or internet connection is needed.

C. Desktop analyzer

An optional Python analyzer reads the serial stream. Every 20 samples it computes three features on a 512-sample (2.56 s) window: normalized line-length (LL) [13], the delta/alpha power ratio (DAR) from a Welch spectrum, and the high-frequency (30–45 Hz) energy ratio (HFE). Each feature has its own adaptive thresholds at $\mu + 2.5\sigma$ and $\mu + 4\sigma$ with fixed floors. A state is proposed when at least two of the three features exceed its threshold, and accepted when 7 of the last 8 analysis ticks agree. Epochs with LL above a ceiling are rejected as artifacts, and adaptation freezes during BLACKOUT. State changes can trigger a web-hook push notification.

D. Companion app

The Android app (Flutter) joins the device's access point, polls `/status` and `/eeg`, and shows the current state, seizure duration, battery level and a live waveform. A simulation endpoint lets caregivers rehearse the alert flow without a seizure.

IV. Evaluation Methodology

A. Data and montage

We used the CHB-MIT Scalp EEG Database [9], [10]: 23 cases from 22 pediatric subjects aged 1.5–22 years, plus case chb24, which was added later. Case chb21 is the subject of chb01, recorded 1.5 years later. All recordings are sampled at 256 Hz with 16-bit resolution in a bipolar 10–20 montage. We included every recording that contains a seizure (191 seizures) plus at least one seizure-free recording per case, for 163 recordings and 223 h in total. The FP1–FP2 derivation Vera measures is not stored directly. We reconstructed it exactly as the signed sum of the bipolar chain FP1-F7, F7-T7, T7-FT9, FT9-FT10, FT10-T8, T8-F8, F8-FP2 wherever a montage contains it (all cases except chb16 and chb17). FP1-F7 and FP2-F8 serve as secondary channels. The data are public and de-identified, and no new human data were collected.

B. Front-end model

Each channel, in μV, passes through a causal model of the analog pass-band: a first-order 0.5-Hz high-pass and a second-order 40-Hz Butterworth low-pass. It is then resampled to 200 Hz with a polyphase anti-aliasing filter, multiplied by the nominal gain of 1000, offset to mid-rail and quantized to 12 bits over 3.3 V. Because the gain of an individual board is uncertain, we repeat the analysis at gains of 500 and 2000.

C. Replaying the detectors

The analyzer is not re-implemented. The published `seizure_monitor.py` is imported unmodified, and its own serial-reader loop is driven by a simulated serial port that emits the firmware's line format, together with a simulated clock that advances 5 ms per line. Warm-up, re-adaptation every 30 s and confirmation delays therefore behave exactly as in live use. Only side effects (plotting, logging, alerts) are stubbed. The firmware's `loop()` is ported line by line to Python and JIT-compiled with Numba, reproducing its integer casts, and its output is the serial stream the analyzer consumes. Each recording starts from a fresh device state.

D. Scoring

Following SzCORE [17], an *alarm* is an entry into an alarm state. We report two definitions: BLACKOUT entries alone, and entries into either BROWNOUT or BLACKOUT. Alarms less than 90 s apart are merged. A seizure counts as detected if an alarm falls within [onset$-$30 s, offset$+$60 s]; every other alarm is a false alarm (FA). We report sensitivity, FA/h and median latency (first alarm minus annotated onset). As a chance reference, a Poisson alarm process with the detector's own alarm rate λ detects a seizure of duration T

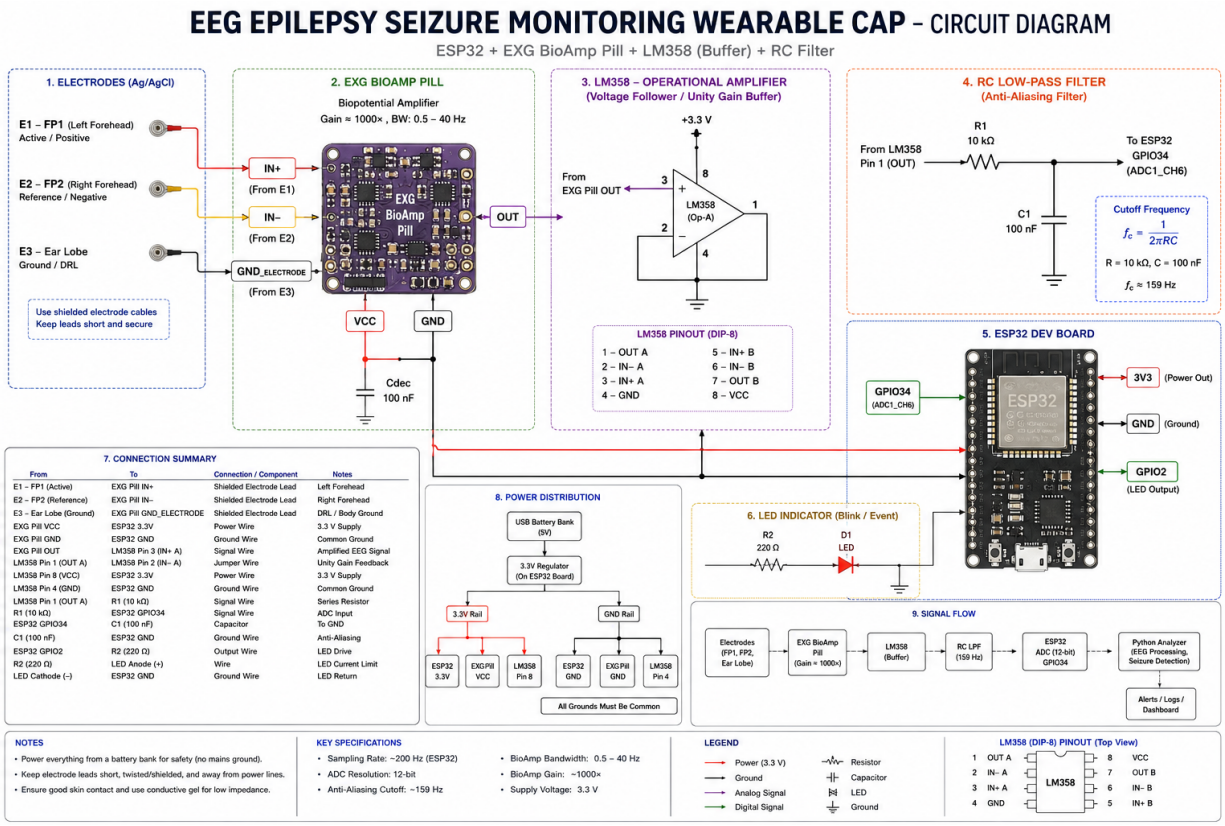


Fig. 1. Vera hardware. Ag/AgCl electrodes at FP1 (+) and FP2 (–), with an earlobe ground/driven-right-leg electrode, feed an EXG BioAmp Pill (gain ≈1000, pass-band 0.5–40 Hz). The output goes through an LM358 voltage follower and an RC low-pass filter (10 kΩ, 100 nF, $f_c \approx 159$ Hz) into the ESP32's 12-bit ADC (GPIO34). The whole system runs from a USB battery bank.

with probability $1 - e^{-\lambda(T+90\,\mathrm{s})}$; we report the mean over seizures.

E. Feature separability

For every analysis tick we logged LL, DAR and HFE, including ticks rejected as artifacts, together with the firmware's 100-ms mean deviation d. We computed the area under the ROC curve (AUC) between ictal ticks and inter-ictal ticks more than 5 min from any seizure.

F. Line-length-ratio detector (LL-R)

Motivated by the feature analysis, we evaluated a deliberately simple, scale-free detector. Let $\ell(t)$ be the analyzer's line-length averaged over each second, and $\beta(t)$ the median of ℓ over the window $[t-330, t-30]$ s. The ratio $R(t) = \ell(t)/\beta(t)$ raises an alarm when $R \geq r$ for D consecutive seconds. A 300-value running median at 1 Hz fits comfortably on the ESP32. The parameters $r \in \{1.5, \ldots, 8\}$ and $D \in \{2, \ldots, 20\}$ s were chosen by patient-wise 2-fold cross-validation (odd versus even case numbers, with chb01 and chb21 kept together). On each training fold we picked the most sensitive (r, D) whose FA/h stayed at or below a target, and we report performance only on the held-out fold.

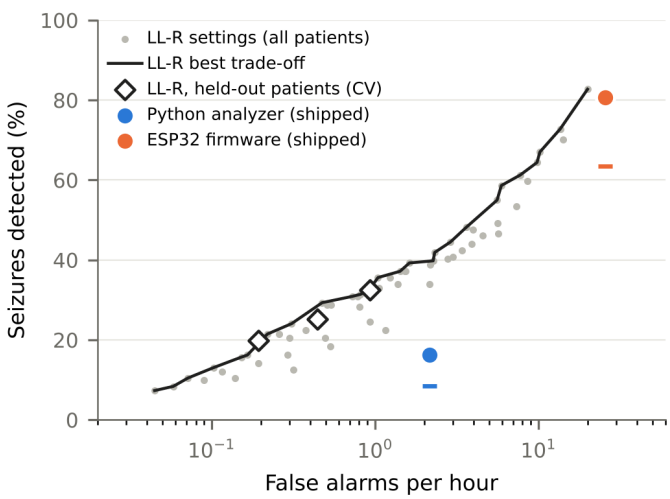


Fig. 2. Sensitivity versus false alarms per hour on FP1-F7. Grey dots are LL-R parameter settings evaluated on all patients (in-sample), and the black line is their best trade-off. Diamonds are LL-R on held-out patients under cross-validation. Filled circles are the two shipped detectors, and the dash below each shows the sensitivity random alarms at the same rate would achieve.

TABLE I
EVENT-BASED RESULTS ON CHB-MIT (FRONT-END MODEL: 0.5–40 HZ, GAIN 1000). "CHANCE" IS THE SENSITIVITY A RANDOM ALARM PROCESS WITH THE SAME ALARM RATE WOULD ACHIEVE.

Channel	Detector / alarm definition	Detected (sens.)	FA/h	Latency (s, median)	Chance
FP1-F7	Python analyzer, BLACKOUT	31/191 (16.2%)	2.16	21.2	8.5%
FP1-F7	Python analyzer, any alarm	82/191 (42.9%)	7.13	19.4	25.1%
FP1-F7	ESP32 firmware, BLACKOUT	154/191 (80.6%)	25.75	25.8	63.5%
FP1-F7	ESP32 firmware, any alarm	178/191 (93.2%)	29.63	22.1	68.4%
FP1-F7	**LL-ratio (proposed, CV)**	48/191 (25.1%)	0.44	27.5	–
FP2-F8	Python analyzer, BLACKOUT	32/191 (16.8%)	2.15	21.0	8.5%
FP2-F8	Python analyzer, any alarm	75/191 (39.3%)	7.04	22.4	24.8%
FP2-F8	ESP32 firmware, BLACKOUT	163/191 (85.3%)	25.94	22.5	63.8%
FP2-F8	ESP32 firmware, any alarm	181/191 (94.8%)	29.71	22.8	68.5%
FP2-F8	**LL-ratio (proposed, CV)**	38/191 (19.9%)	0.47	23.5	–

V. RESULTS

A. Shipped detectors

Table I summarizes the results. On FP1-F7, the analyzer's BLACKOUT alarm detected 31/191 seizures (16.2%) at 2.16 FA/h (51.9 per day), with a median latency of 21.2 s. Counting BROWNOUT warnings as alarms raised sensitivity to 42.9% but also raised FA/h to 7.13. The firmware detector entered BLACKOUT many times per hour (25.75 FA/h). Its sensitivity of 80.6% is close to the 63.5% that random alarms at the same rate would achieve, so most of its detections are coincidental. The cause is visible in its design: with a forgetting factor of 0.995 per sample, its spread estimate has a memory of about 1 s, so a 4σ excursion lasting 75 ms (15 samples) is common in ordinary EEG.

B. Which features carry the seizure

Line-length was the only analyzer feature with useful separation (pooled AUC 0.65, median per-patient AUC 0.79; Fig. 3). The firmware deviation d behaved similarly (AUC 0.65, per patient 0.77). The gap between the pooled and per-patient AUCs shows that baseline line-length varies widely between patients, so a fixed threshold cannot suit everyone, while a threshold relative to each patient's own baseline can. DAR (0.45) and HFE (0.48) were at or below chance. On a frontal channel, inter-ictal EEG is rich in slow eye-movement activity, which likely inflates delta power outside seizures. The 30–45-Hz band carries little energy in these mostly rhythmic, low-frequency pediatric seizures. Because the analyzer requires two of its three features to vote together, these two weak features suppress line-length. LL's fixed floors (300 and 600 ADC units) add a further problem: they are absolute values whose meaning depends on the front-end gain, and the median ictal LL was 163 units. Artifact rejection discarded 1.65% of ictal and 2.714% of inter-ictal ticks.

C. Line-length-ratio detector

With the training-fold target set to 1 FA/h, LL-R detected 62/191 seizures (32.5%) in held-out patients at 0.93 FA/h, with a median latency of 22.0 s. Stricter targets traded sensitivity for fewer false alarms: 25.1% at 0.44 FA/h and 19.9% at 0.19 FA/h (Table II, Fig. 2). Because R is a ratio, LL-R is unaffected by the front-end gain. The shipped detectors are not (Table III).

TABLE II
LL-R UNDER PATIENT-WISE 2-FOLD CROSS-VALIDATION (HELD-OUT FOLDS ONLY). PARAMETERS ARE r/D CHOSEN ON EACH TRAINING FOLD.

Ch.	FA/h target	r/D (folds)	Detected	FA/h	Lat. (s)
FP1-F7	$\leq$1.0	4.0/8; 3.0/10	62/191 (32.5%)	0.93	22.0
FP1-F7	$\leq$0.5	3.0/20; 4.0/10	48/191 (25.1%)	0.44	27.5
FP1-F7	$\leq$0.2	4.0/20; 3.5/20	38/191 (19.9%)	0.19	32.0
FP2-F8	$\leq$1.0	2.5/20; 3.0/10	56/191 (29.3%)	1.02	28.0
FP2-F8	$\leq$0.5	4.0/15; 5.0/8	38/191 (19.9%)	0.47	23.5
FP2-F8	$\leq$0.2	5.0/20; 8.0/8	22/191 (11.5%)	0.14	26.0

TABLE III
SENSITIVITY / FA/H ON FP1-F7 AS THE ASSUMED FRONT-END GAIN VARIES (BLACKOUT ALARMS; LL-R AT THE $\leq$0.5 FA/H CV TARGET).

Gain	Analyzer	Firmware	LL-R (CV)
$\times$500	8.4% / 0.60	70.2% / 18.65	25.1% / 0.44
$\times$1000	16.2% / 2.16	80.6% / 25.75	25.1% / 0.44
$\times$2000	28.3% / 4.14	90.1% / 28.46	25.1% / 0.44

VI. DISCUSSION AND LIMITATIONS

The benchmark changes how Vera's alarms should be described. The on-device detector, as shipped, mostly raises alarms at random, so its detections largely reflect how often it fires, and it should not be presented as seizure detection. The analyzer's "pre-ictal" BROWNOUT state is better described as a low-confidence warning: nothing in its design or in our results shows that it predicts seizures. A simple, scale-free line-length rule does better on held-out patients and is cheap enough to run on the device, so we are moving it into the firmware.

The absolute performance also needs saying plainly. Even the best single-channel detector here catches only about a third of seizures at roughly one false alarm per hour, well short of what clinical guidelines expect from a seizure alarm [4]. Per-patient results vary widely (Fig. 4). Patient-specific calibration and a second channel (e.g., a temporal electrode) are the clearest routes to improvement; multichannel, patient-

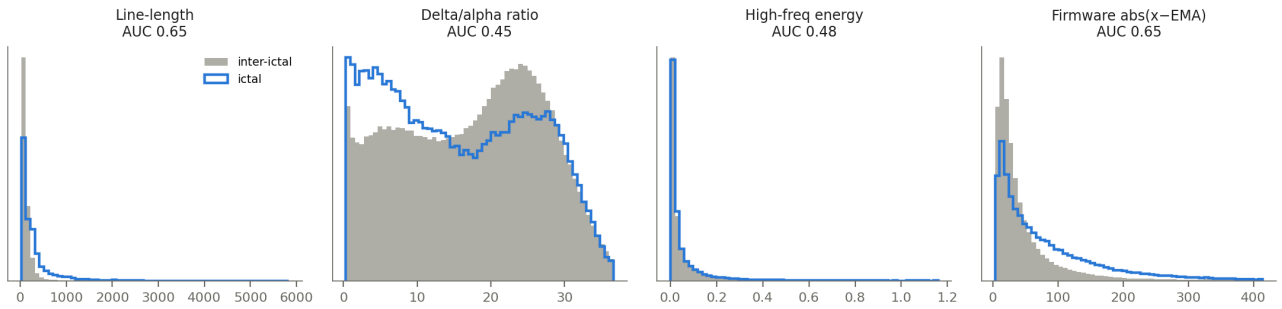


Fig. 3. Distributions of each feature on FP1-F7, ictal (blue outline) versus inter-ictal (grey), with the pooled AUC.

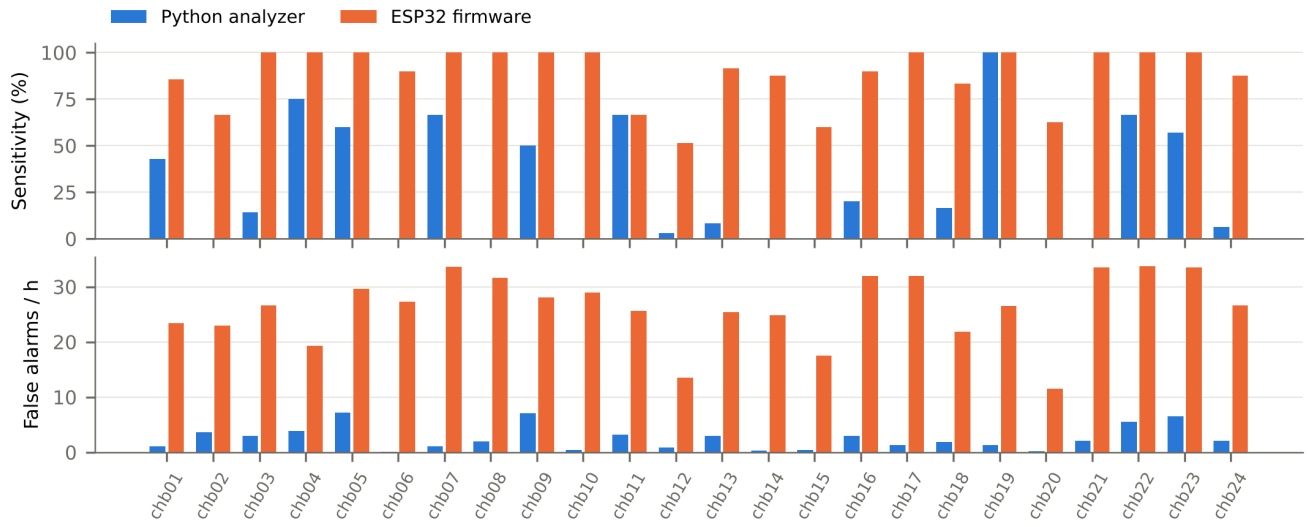


Fig. 4. Per-patient sensitivity (top) and false alarms per hour (bottom) of the two shipped detectors (BLACKOUT alarms, FP1-F7).

specific classifiers reach much higher sensitivity on the same database [14].

The study has several limitations. (i) CHB-MIT is a pediatric, mostly focal-onset dataset recorded with clinical amplifiers, so adult, ambulatory and gel-free recordings will be noisier, especially on the forehead, where eye and muscle artifacts dominate. (ii) The front-end is simulated, so real electrode impedance, motion, 50-Hz mains interference and the 159-Hz RC filter's weak anti-aliasing at 200 Hz sampling are not modelled. (iii) Each recording starts cold, so seizures in the first minutes of a file are harder to detect than in continuous use. (iv) One frontal channel cannot localize seizures, and posterior or temporal seizures may be invisible on it. (v) The firmware Wi-Fi access point uses a fixed password, which would need to change before any unattended deployment. Vera is a research prototype and not a medical device. Bench characterization of the hardware (input-referred noise [TODO: μV_{rms}], battery life [TODO: h]) and a supervised recording study under ethics approval are the next steps.

VII. Conclusion

We presented Vera, an open, low-cost single-channel EEG seizure monitor, together with a reproducible benchmark that replays public, expert-annotated EEG through a model of the device and runs its unmodified detector code. The benchmark showed that line-length, not the spectral features, carries the seizure information on a frontal channel, and that the shipped detectors are limited by short-memory adaptation, a vote that lets weak features override strong ones, and fixed, gain-dependent thresholds. A cross-validated line-length-ratio detector that fits on the ESP32 doubled the analyzer's sensitivity at a lower false-alarm rate on held-out patients. Absolute single-channel performance remains modest, and we report it as such. We release the hardware, firmware, analyzer and benchmark so that other low-cost monitors can be evaluated the same way.

Data and Code Availability

CHB-MIT is available from PhysioNet [10]. All Vera hardware files, firmware, the analyzer, the app and the evaluation notebook are available at [18].

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evaluation code and draft the manuscript; the authors reviewed all content and take full responsibility for it.

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